

Beyond Symbolic Mind: Re-evaluating the Logical Model of Intelligence

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Abstract

Symbolic logic has long been treated as a model of intelligence, motivating rule-based inference and classical approaches to artificial intelligence. Yet systems built on explicit symbolic derivation have proven difficult to optimize, brittle under variation, and unable to generalize beyond tightly specified domains. These limitations are not merely engineering obstacles. The validity of long-chain symbolic inference depends on conditions that the world rarely affords: each step requires jointly realizable states and transition relations that must remain constructible throughout the chain. When any such condition cannot be instantiated, collapse is systemic rather than local. Human cognition, by contrast, operates under partial information and incomplete state specification, relying on pattern matching rather than the construction and maintenance of fully specified symbolic derivations. This work re-evaluates the long-standing assumption that logical form constitutes the operative architecture of intelligence and clarifies why symbolic inference, though formally coherent, cannot serve as an operative account of cognition.

Keywords: Logical Inference, Symbolic Computation, Symbolic Grounding, Human Intelligence

1 Introduction

For much of intellectual history, logical inference and symbolic computation have been treated as the emblem of rationality. From early analytic philosophy to formal AI, intelligence was equated with the capacity to manipulate explicit propositions through rule-governed operations [1–3]. In this view, reasoning is a procedure of syntactic precision: the more faithful the derivation, the closer it approaches ideal cognition.

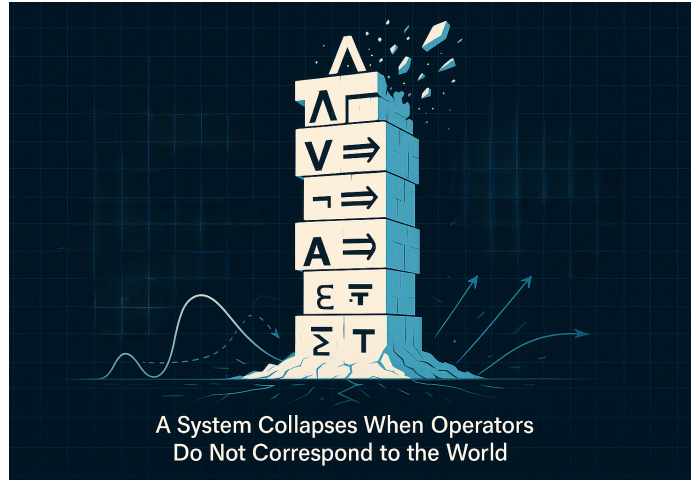


Fig. 1 Symbolic inference collapses under two conditions: (i) conjunctive states in a long-chain derivation fail to exist, and (ii) operators lack generative grounding. Formal validity alone therefore cannot ensure real-world validity.

Yet across both AI development and cognitive practice, symbolic systems have proven notoriously brittle. Their optimization demands exhaustive specification, their generalization remains narrow, and their operational success depends on conditions far more complete than those available in realistic environments. Human cognition itself does not resemble such systems: everyday reasoning does not proceed by enumerating formal operators but by navigating incomplete, ambiguous, and partially grounded worlds. The purity attributed to symbolic computation is largely a product of notation rather than behavior.

The difficulty is not merely computational cost but ontological feasibility. A symbolic chain is valid only if every intermediate state is jointly realizable and every operator preserves a constructible transition in the world. When even one condition fails to correspond to how states and transitions can actually be realized, the breakdown is not localized but systemic. Classical logic assumes that symbols remain stable under inference; real environments do not. What appears formally consistent at the syntactic level may therefore be semantically impossible at the level of world grounding.

This work revisits the grounding conditions presupposed by symbolic inference and makes explicit why those conditions rarely hold outside idealized abstraction. In doing so, we argue that intelligence cannot be defined as symbolic manipulation. The validity of inference depends not on internal closure, but on whether its states and operations correspond to realizable states and transitions in the world.

Our contribution. This work identifies the grounding conditions under which logical inference and symbolic computation can apply beyond formal abstraction. Their central vulnerability is ontological rather than computational: inference requires jointly realizable states and world-valid operator transitions, and fails when these conditions are not met. Drawing on behavioral and cognitive evidence, we further show that

human reasoning does not operate as formal symbolic inference, but relies on pattern matching. The grounding analysis explains why symbolic computation can remain formally coherent yet empirically fragile, while the cognitive evidence shows why symbolic inference cannot provide a general account of intelligence.

2 Two Grounding Conditions for Logical Entailment Beyond Abstraction

Although widely framed as issues of scalability or computational expansion, the core difficulty of symbolic inference arises at the level of definability rather than implementation. For multi-step entailment to count as entailment, two structural premises must hold simultaneously. These are the conditions that make rule-governed inference possible:

1. Joint Specifiability.

Distinct propositions must admit a coherent joint state within the same formal representational space ($A \wedge B$ must be definable, not merely individually well-formed).

2. Determinable Transition.

Successive symbolic states must be linked by an operator-governed rule of transformation, rather than by formal succession alone ($A \rightarrow B$ must correspond to a determinable derivation rule).

These premises rarely appear explicitly because they function as background commitments rather than articulated conditions. The following sections clarify how they operate as the very terms under which symbolic entailment is possible across extended derivations.

Table 1 Constructibility Failure Modes of Symbolic Inference

Formal Operation	Syntactic Status	Ontic Failure Condition
$p \rightarrow q$	valid	q not uniquely realizable by p
$\neg q \rightarrow \neg p$ (contraposition)	valid	absence of q does not imply absence of p
$p \wedge r$ (conjunction)	valid	joint state not constructible in the world
$\neg p$ (global negation)	valid	world lacks generative operator for full complement
$p \rightarrow q \rightarrow r \rightarrow s$	valid	any non-admissible step collapses entire chain

2.1 Event Co-Realizability: When Joint Propositions Must Exist in the World

For symbolic inference to extend beyond a single step, distinct propositional states must admit a jointly realizable configuration within the same formal space. That is, if A and B are to appear in an entailment sequence, then $A \wedge B$ must constitute an admissible state rather than a merely formal conjunction. This requirement concerns

joint realizability: propositions must be not only individually well-formed but capable of standing together within a single derivational framework.

When this joint condition is not met, the resulting expressions may remain syntactically valid while lacking any realizable state within the inferential space. In such instances, extended entailment continues to proceed formally, yet does so without corresponding instantiation.

2.2 Operational Constructibility: When Logical Operators Must Correspond to World-Valid Transitions

A second condition for extending symbolic inference beyond abstraction is that represented states be connected by a determinate operation. In an entailment sequence, $A \rightarrow B$ must specify more than syntactic succession: there must be an operation that accounts for how the state represented by A is transformed into the state represented by B . Beyond a purely formal system, that operation must describe a transition that can actually occur under the relevant conditions in the world.

When such constructibility is absent, expressions may remain syntactically permissible while no determinate transformation can be assigned to connect A with B . In these instances, derivation proceeds formally but without an operational pathway.

3 The Accumulation Problem in Long-Chain Derivation

Within symbolic traditions, the capacity for extended inference has often been treated as the defining marker of rational competence. Yet the feasibility of such extension rests not on computational endurance but on the continued instantiability of each inferential step. A derivation requires that successive propositions remain jointly assignable and that the transition between them remain operator-governed. As the chain extends, these conditions do not merely increase in number; they compound.

Failure at any point does not merely impair the subsequent step but collapses the entire structure of admissibility: a single non-instantiable transition eliminates the possibility of a world in which the chain can hold at all. What remains is syntactic form without feasible grounding. Long-chain derivation thus encounters a structural accumulation problem: the conditions required for joint admissibility and operator constructibility become increasingly restrictive with each added step. What appears as inferential continuity at the syntactic level may therefore cease to be constructible at the level of world applicability.

Intuitive Motivation. When we reason step by step, it is easy to assume that if each part works, the whole chain should work too. In practice, however, the conditions invoked by different steps cannot always be satisfied together. As a chain grows, its requirements accumulate faster than they can be jointly upheld. The issue is not error but the simple fact that extended inference rarely remains fully instantiable.

4 Why Human Cognition Rarely Executes Formal Inference

The difficulty of long-chain derivation, previously established as a matter of instantiability rather than computational growth, directly entails that human cognition seldom realizes such forms. To execute formal inference, states must remain jointly assignable and operator-governed across successive steps. Yet ordinary cognition is seldom situated in informational conditions that deliver complete state specification. Most situations present partial, intermittent, or non-exhaustive data, and memory does not retain fully articulated intermediate configurations across extended sequences. In such environments, long-chain derivation does not arise as a practical mode of cognition because the information and memory capacity required to sustain it are rarely available.

Consistent with this analysis, substantial behavioral evidence shows that human reasoning relies on pattern matching rather than formal symbolic inference. Findings on heuristics [4], post-hoc fitting [5], and representation bias [6] provide convergent evidence for this account of human reasoning.

Intuitive Motivation. Long chains of logical inference are rare not because humans fail at logic, but because real situations seldom provide complete information or enough memory to hold every intermediate state at once.

4.1 Mathematical and Logical Reasoning as Pattern Matching

Mathematical and logical reasoning are commonly treated as paradigmatic cases of formal symbolic inference. Their rules and standards of correctness are explicitly specified, so if human cognition reliably executed symbolic procedures, it should be most evident in these domains. Yet unaided mathematical performance remains both error-prone and unstable. Mathematical errors are not merely random lapses: people repeatedly misapply familiar operations, lose intermediate relations, or extend a locally recognized pattern beyond the conditions under which it holds. Even expert proofs require repeated checking and correction. Such structured instability is more consistent with the local matching and patching of familiar forms than with the reliable execution of invariant formal rules.

Logical reasoning also exhibits the same pattern. Numerous systematic departures from formal logical inference can be explained by reliance on pattern matching. Findings on heuristic reasoning [4], post-hoc fitting [5], and representativeness bias [6] show that responses are shaped by familiar patterns and salient cues rather than by the construction and maintenance of explicit symbolic derivations. Correct and incorrect conclusions may therefore arise from the same pattern-matching process, depending on whether the matched pattern aligns with the formally valid result.

4.2 Pattern Matching Can Produce Rule-Like Behavior

Pattern matching can produce behavior that appears rule-governed without requiring the explicit representation or execution of a formal rule. When a familiar configuration reliably maps onto the same response, repeated matching generates stable and systematic outputs. If the matched pattern aligns with a valid mathematical or logical relation, the resulting performance is indistinguishable from rule-following at the level of the observed answer.

This equivalence breaks down when familiar patterns no longer determine the correct response. Changes in surface form, novel combinations, or longer sequences can preserve the underlying formal rule while disrupting the cues on which pattern matching depends. Correct and incorrect responses can therefore arise from the same cognitive process: performance appears rule-governed when the available pattern aligns with the formal solution, and becomes predictably unstable when it does not.

5 Why Symbolic Operations Do Not Constitute Intelligence

Symbolic inference can apply to the world only when its grounding conditions are already satisfied: the represented states must be jointly realizable, and the operators must correspond to constructible transitions. Under such completeness, the informational gain of symbolic chaining becomes narrowly combinatorial—it yields rearrangements of known constraints, not expansion into genuinely uncertain space. The system can produce new forms of expression, but only within the finite closure already guaranteed by its premises. Therefore, intelligence cannot be defined as a mode of operation whose applicability to the world depends on these conditions having already been established.

In contrast, pattern-based architectures operate under partial information. They do not require exhaustive specification of every predicate or causal condition; instead, they generalize from limited, structurally adjacent evidence to unseen regimes. Their success is not accidental looseness but a functional property: generalization requires tolerance of missing constraints, not their full satisfaction. Where symbolic methods collapse once a single grounding fails, pattern systems degrade gracefully, still producing usable approximations.

6 Discussion

6.1 Formal Validity Does Not Establish Semantic Grounding

Within the formal mathematical tradition, well-formed definitions and valid derivations establish what follows within an abstract system. They do not establish that the states and operations represented by that system are realizable in a target domain.

Accordingly, applying mathematics to a target domain requires semantic grounding: formal states must correspond to realizable states, and formal operators to world-valid transitions. If either correspondence fails, further derivation may remain

formally correct but cannot support claims about that domain. Formal rigor establishes consequences within a system; it does not establish that the system applies to the world.

6.2 Intelligence as World-Grounded Constructibility

Intelligence is often equated with expanding computational capacity—greater logical depth, representational precision, or algorithmic scale. The present account clarifies a different boundary: inference cannot surpass the grounding of its operators in the world. When one step in a derivation lacks correspondence to how things actually operate, the failure is systemic rather than local, because formal validity remains indifferent to real constraints.

True intelligence requires grounding in the conditions it aims to describe. Computation or simulation, however extensive, is insufficient when it lacks alignment with the mechanisms it invokes. Grounding is not an accessory but the basic condition that prevents intelligence from devolving into internally coherent yet externally irrelevant formulation.

6.3 Externalization as the Condition for Extended Inference

Extended logical inference exceeds what human cognition can internally maintain. As chains lengthen, intermediate states outstrip both available information and memory capacity. For this reason, long-form inference is rarely executed in isolation and requires continual calibration through peers, texts, and formal expertise. The same limitation explains why we write steps down, segment reasoning, and externalize intermediate states; without such stabilization, extended inference does not hold.

7 Conclusion

This work examined the feasibility conditions required for symbolic inference to function beyond formal isolation. Systems that rely on explicit derivational chains must presuppose that each intermediate state remains jointly realizable and that every transition relation is constructible in the world. We showed that these requirements, rather than computational cost alone, underlie the brittleness, limited generalization, and optimization difficulty widely observed in symbolic approaches. When a single condition cannot be instantiated, failure is not incremental but global, eliminating the possibility of a world in which the chain can hold.

Human cognition, by contrast, does not generally operate by constructing and maintaining fully specified symbolic derivations. Behavioral evidence indicates that reasoning relies on pattern matching, which can produce rule-like performance when familiar patterns align with formally valid results, but predictable errors when they do not. The historical identification of intelligence with symbolic inference is therefore misplaced. Symbolic inference can derive consequences once its states and operators have been grounded, but it cannot itself establish the conditions that make those derivations applicable to the world.

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Data and code availability

No datasets were generated or analysed, and no code was developed or used in this study.

Author contributions

E.D. conceived the study, developed the central argument, and wrote and revised the manuscript.

If any of the sections are not relevant to your manuscript, please include the heading and write ‘Not applicable’ for that section.

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